

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

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Abstract

This project analyzes product line profitability and margin performance for Nassau Candy Distributor using data analytics and interactive business intelligence techniques. An interactive dashboard was developed using Python, Pandas, Plotly, and Streamlit to evaluate product profitability, gross margin performance, revenue concentration, divisional efficiency, and cost optimization opportunities. The analysis identified high-profit product lines, margin improvement opportunities, revenue dependency risks, and underperforming products requiring strategic review. The project provides actionable insights to support pricing decisions, product portfolio optimization, supplier negotiations, and long-term profitability growth.

1. Introduction

Nassau Candy Distributor manages a large portfolio of confectionery and specialty food products across multiple product divisions and geographic markets. While revenue growth is an important business objective, sales volume alone does not accurately reflect financial performance. Some products generate substantial revenue but contribute limited profit due to low margins, while others generate strong profitability despite lower sales volumes.

Without a structured profitability analysis framework, it becomes difficult to identify the products, divisions, and geographic markets that drive business success. This project was developed to transform transactional sales and cost data into actionable business intelligence, enabling management to evaluate profitability performance, optimize product portfolios, improve margin efficiency, and support data-driven decision-making.

2. Objectives

The main objectives of the project were:

- Analyze product-level profitability performance.

- Evaluate gross margin trends across products and divisions.
- Identify high-profit and high-margin products.
- Detect high-sales but low-margin products.
- Assess divisional revenue and profitability performance.
- Measure revenue and profit concentration risk.
- Identify cost optimization opportunities.
- Develop an interactive profitability analytics dashboard.

3. Dataset Description

The dataset contains product sales, costs, and profitability information across multiple product categories and business divisions.

Table 1. Dataset Fields

Field	Description
Product Name	Product description
Division	Business division
State/Province	Geographic sales region
Sales	Total revenue generated
Gross Profit	Total profit after cost deduction
Gross Margin %	Profitability percentage
Cost	Product cost

The dataset was used to analyze product profitability, margin performance, revenue concentration, divisional efficiency, and cost structure behavior.

4. Methodology

The project followed a structured analytical workflow to transform raw sales and cost data into meaningful profitability insights. Initial preprocessing involved data cleaning, validation of financial records, handling missing values, and verifying sales, cost, and gross profit calculations to improve data quality and consistency.

Following preprocessing, profitability metrics such as Gross Margin Percentage and Profit Conversion Efficiency were analyzed to evaluate product and division performance. Product-level, division-level, geographic, and cost structure analyses were conducted to identify high-profit products, margin improvement opportunities, revenue concentration risks, and underperforming product lines. An interactive dashboard was developed using Python, Pandas, Plotly, and Streamlit to enable KPI monitoring, profitability analysis, and data-driven business decision-making.

5. Feature Engineering

The raw dataset was enhanced by creating additional business metrics to support profitability and performance analysis. Gross Margin Percentage was calculated to measure how efficiently sales revenue was converted into profit, while Profit Per Unit was created to evaluate profitability at the product level. Month and Year features were extracted from the Order Date field to support monthly and yearly trend analysis. These engineered features enabled deeper insights into product profitability, margin performance, revenue trends, and operational efficiency across the business.

Table 2. Engineered Features

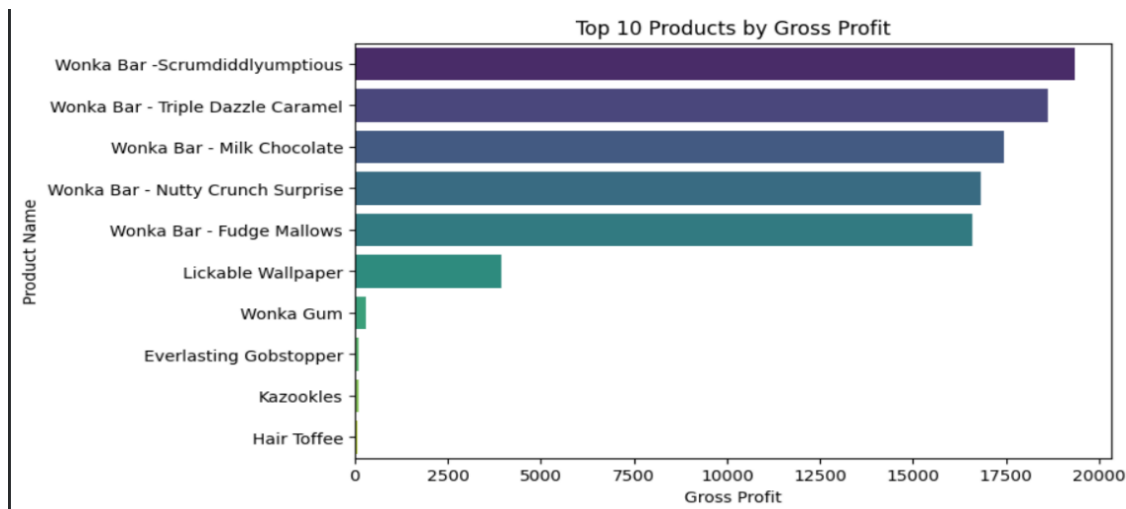
Feature	Description
Gross Margin %	Measures profitability relative to sales revenue
Profit Per Unit	Profit generated for each unit sold
Month	Extracted from Order Date for monthly trend analysis
Year	Extracted from Order Date for yearly performance analysis

The engineered features enhanced profitability analysis and enabled deeper insights into product performance, margin efficiency, and revenue trends.

6. Exploratory Data Analysis and Insights

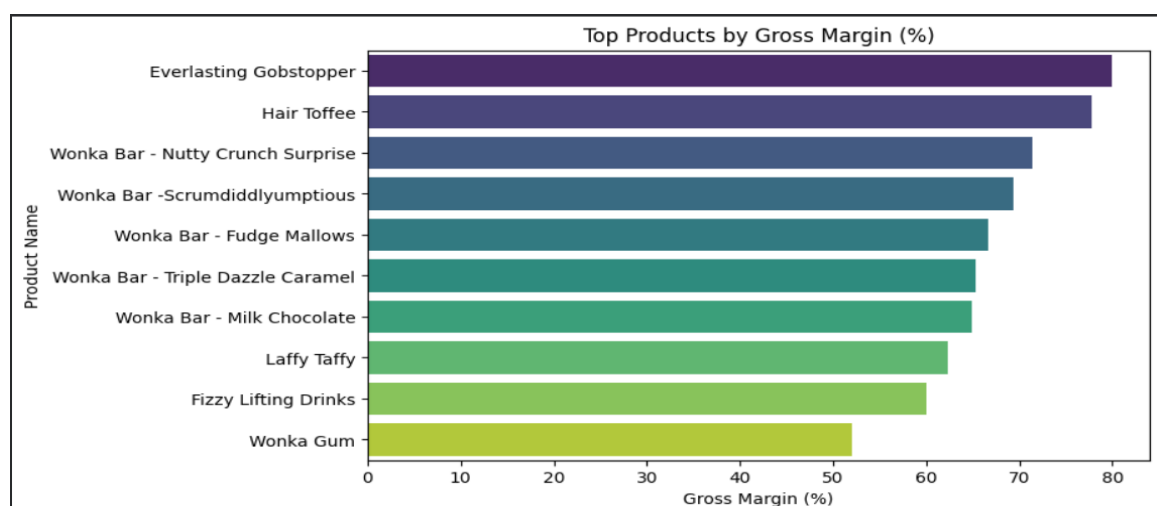
Exploratory Data Analysis (EDA) was performed to understand product profitability, gross margin performance, revenue concentration, divisional efficiency, and cost structure behavior within Nassau Candy Distributor. Various visualizations were developed to identify high-performing products, margin improvement opportunities, revenue dependency risks, and underperforming product lines. The insights obtained from this analysis support data-driven decision-making for pricing strategy, product portfolio optimization, supplier negotiations, and overall profitability improvement.

1. Top Products by Gross Profit Analysis



The gross profit analysis identifies the products contributing the highest profitability to Nassau Candy Distributor. The Wonka Bar product line dominates profit generation, with Scrumdiddlyumptious, Triple Dazzle Caramel, Milk Chocolate, Nutty Crunch Surprise, and Fudge Mallows emerging as the top-performing products. A significant gap exists between these products and the remaining items, indicating that a relatively small group of products contributes a substantial portion of total profit. This concentration highlights the strategic importance of maintaining strong inventory availability, pricing strategies, and promotional efforts for these high-performing products to sustain overall business profitability.

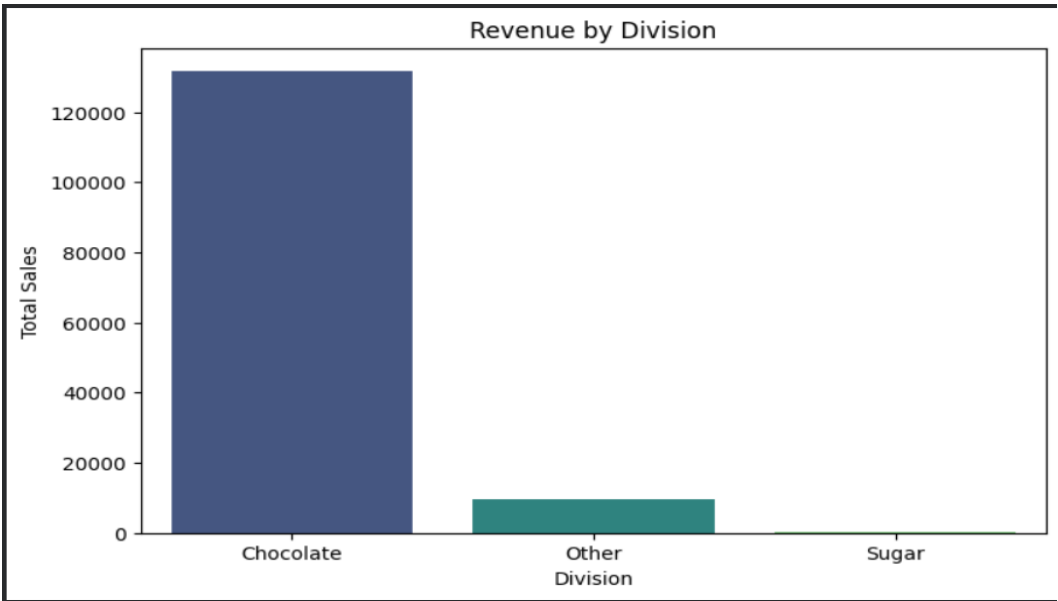
2. Top Products by Gross Margin (%) Analysis



The gross margin analysis identifies the products that generate the highest profit relative to their sales revenue. Everlasting Gobstopper and Hair Toffee achieve the strongest margin performance, indicating exceptional profitability and efficient cost

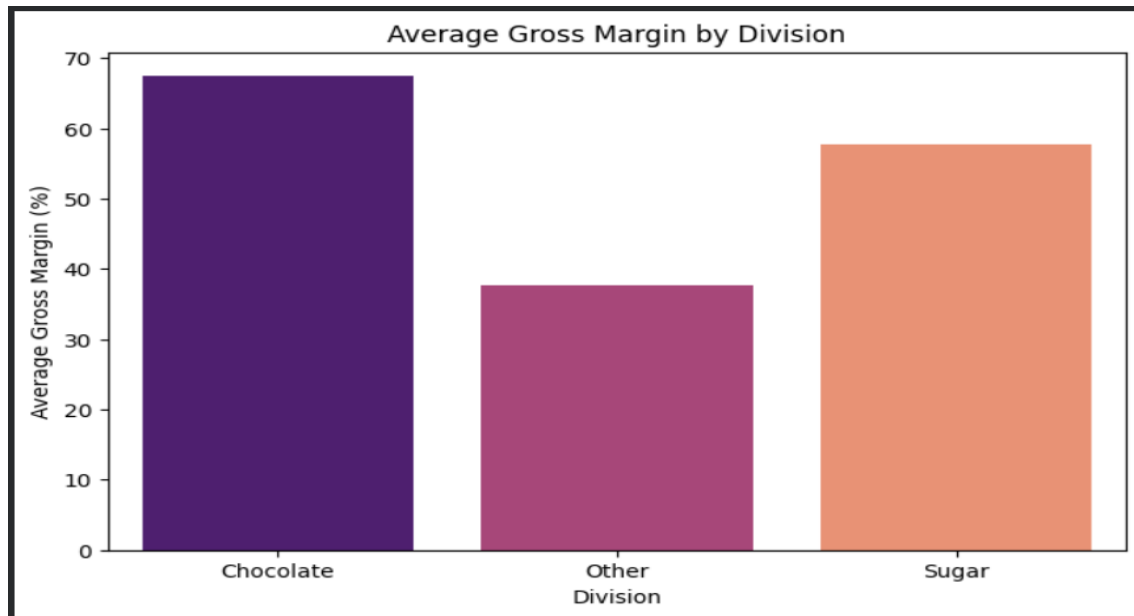
management. Several Wonka Bar products, including Nutty Crunch Surprise, Scrumdiddlyumptious, and Fudge Mallows, also maintain strong margin percentages, demonstrating their ability to convert revenue into profit effectively. The presence of both high-margin niche products and popular Wonka Bar products suggests a balanced portfolio that supports profitability through both efficiency and sales volume. These findings can help management prioritize high-margin products in pricing, marketing, and inventory strategies to maximize overall business performance.

3. Revenue Performance by Division Analysis



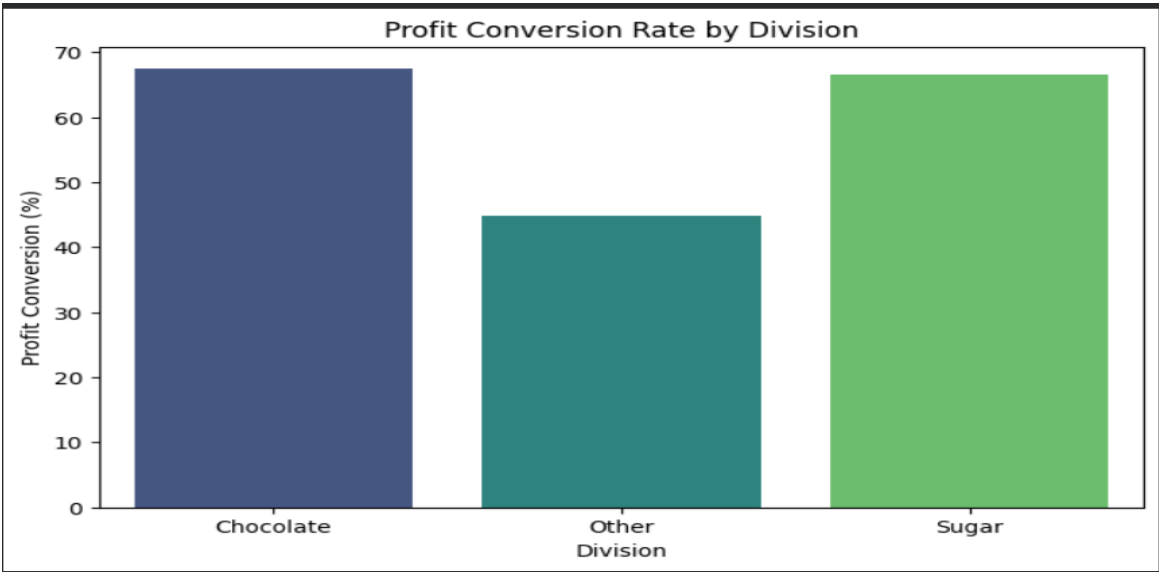
The division-level revenue analysis reveals that the Chocolate division overwhelmingly dominates Nassau Candy's sales performance, generating significantly higher revenue than both the Other and Sugar divisions. The substantial gap between Chocolate and the remaining divisions indicates that the company's revenue is highly concentrated within chocolate-based products. While this highlights the strong market demand and commercial success of the Chocolate division, it also suggests a degree of dependency on a single product category. Expanding the performance of other divisions and diversifying revenue sources could help reduce business risk while supporting more balanced long-term growth.

4. Average Gross Margin by Division Analysis



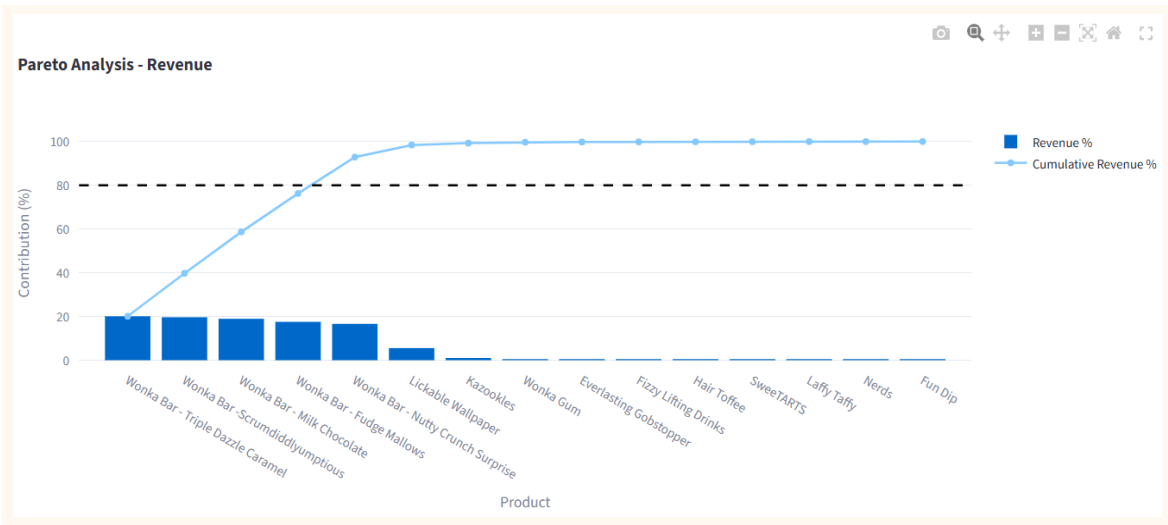
The average gross margin analysis evaluates how efficiently each division converts sales revenue into profit. The Chocolate division achieves the highest average gross margin, indicating strong profitability and effective cost management. The Sugar division also demonstrates healthy margin performance, while the Other division records the lowest margin percentage among all divisions. Although the Chocolate division generates the highest revenue, it also maintains superior profit efficiency, making it the strongest-performing division overall. These findings suggest that management should continue investing in high-margin divisions while exploring opportunities to improve profitability within lower-margin product categories.

5. Profit Conversion Efficiency by Division Analysis



The profit conversion efficiency analysis measures how effectively each division converts sales revenue into gross profit. The Chocolate division achieves the highest profit conversion rate, closely followed by the Sugar division, indicating strong profitability and efficient cost management. In contrast, the Other division records a considerably lower conversion rate, suggesting that a larger portion of its revenue is consumed by costs. The similarity between the Chocolate and Sugar divisions demonstrates their ability to consistently generate profit from sales, while the performance gap highlights opportunities to improve efficiency within the Other division. These findings provide valuable insights for resource allocation, pricing strategies, and profitability optimization across the business.

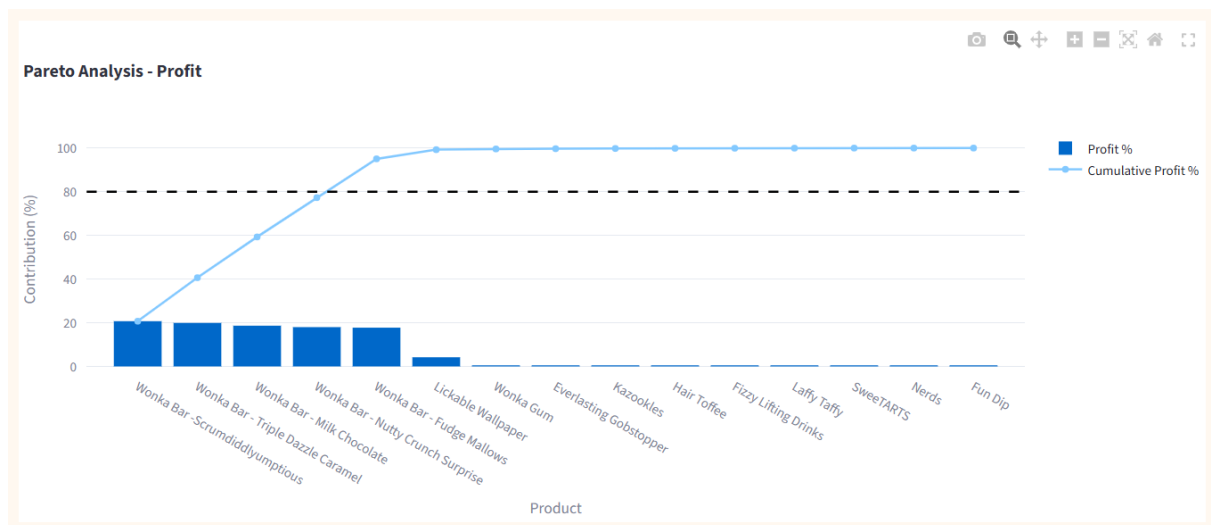
6. Revenue Concentration (Pareto) Analysis



The Pareto analysis reveals a strong revenue concentration within Nassau Candy Distributor’s product portfolio. Only **26.67% of products (4 out of 15 products)**

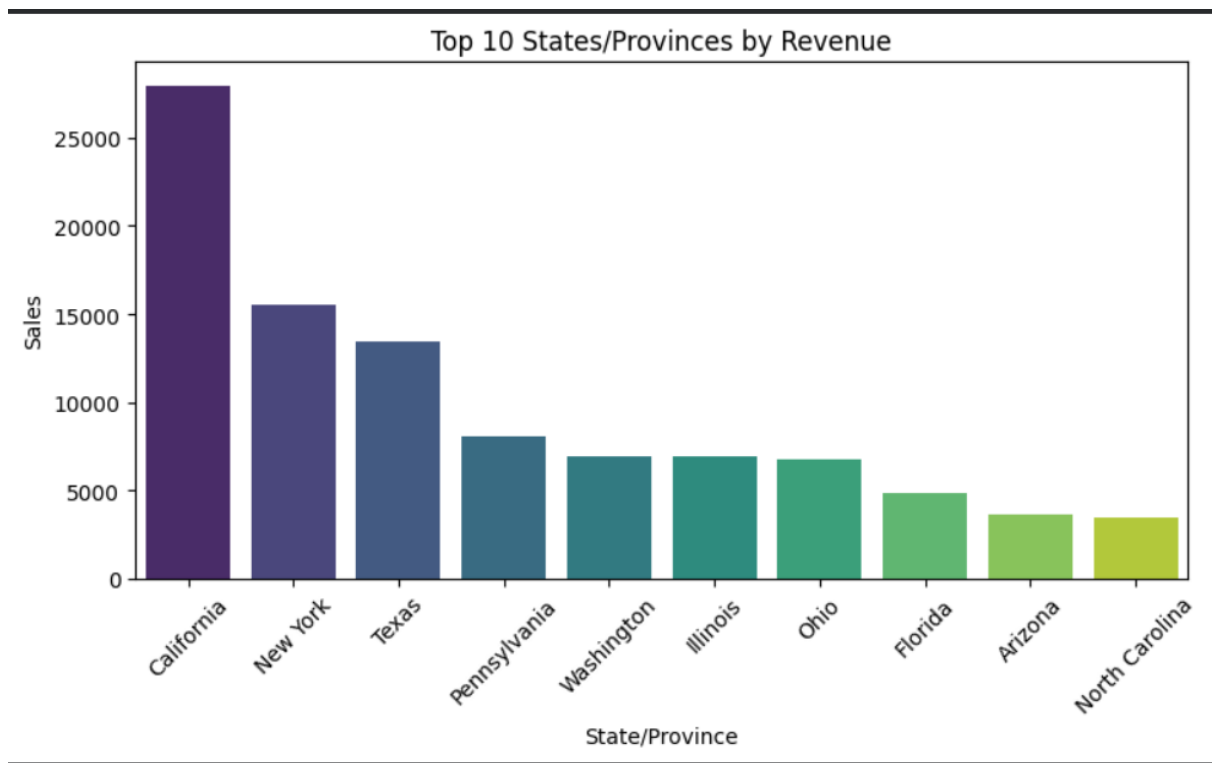
generate approximately **80% of total revenue**, indicating a classic 80/20 business pattern. The leading contributors are **Wonka Bar – Triple Dazzle Caramel, Wonka Bar – Scrumdiddlyumptious, Wonka Bar – Milk Chocolate, and Wonka Bar – Fudge Mallows**, which collectively account for the majority of sales revenue. Revenue contribution declines sharply after these top-performing products, while the remaining products contribute relatively little to overall revenue. These findings highlight the company's dependence on a small group of high-performing products and emphasize the importance of maintaining inventory availability, pricing competitiveness, and marketing focus for these key revenue drivers.

7. Profit Concentration (Pareto) Analysis



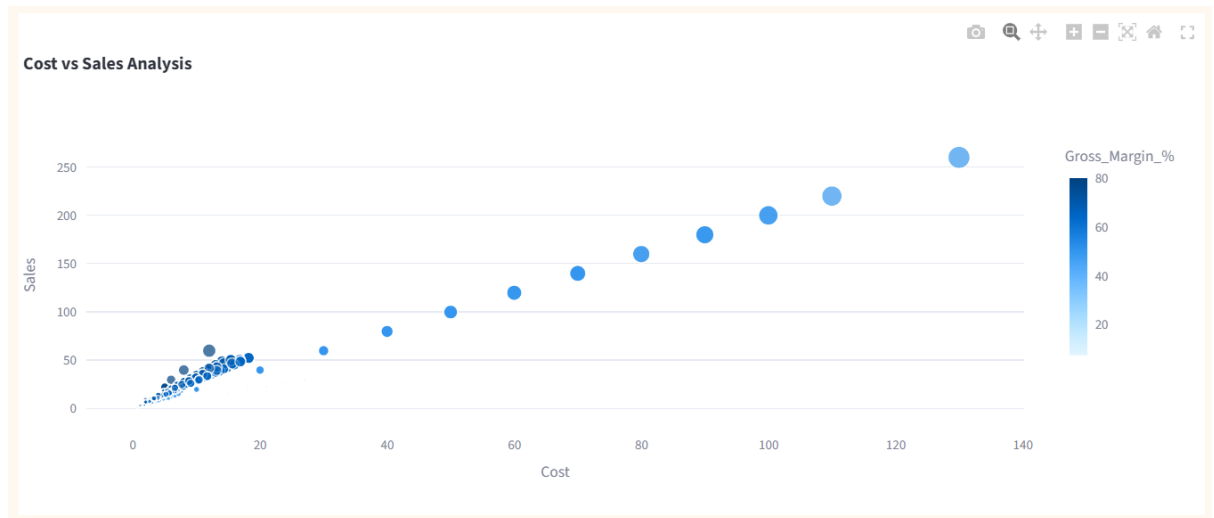
The Pareto analysis of gross profit reveals that profit generation is highly concentrated within a small number of products. Only **26.67% of products (4 out of 15 products)** contribute approximately **80% of total gross profit**, demonstrating a strong profit concentration pattern. The primary profit drivers are **Wonka Bar – Scrumdiddlyumptious, Wonka Bar – Triple Dazzle Caramel, Wonka Bar – Milk Chocolate, and Wonka Bar – Nutty Crunch Surprise**, which collectively account for the majority of profitability. After these products, profit contribution declines sharply, indicating that the remaining products have a relatively limited impact on overall earnings. This finding highlights the strategic importance of protecting and expanding the performance of these key profit-generating products through effective pricing, inventory management, and targeted marketing initiatives.

8. Regional Revenue Concentration Analysis



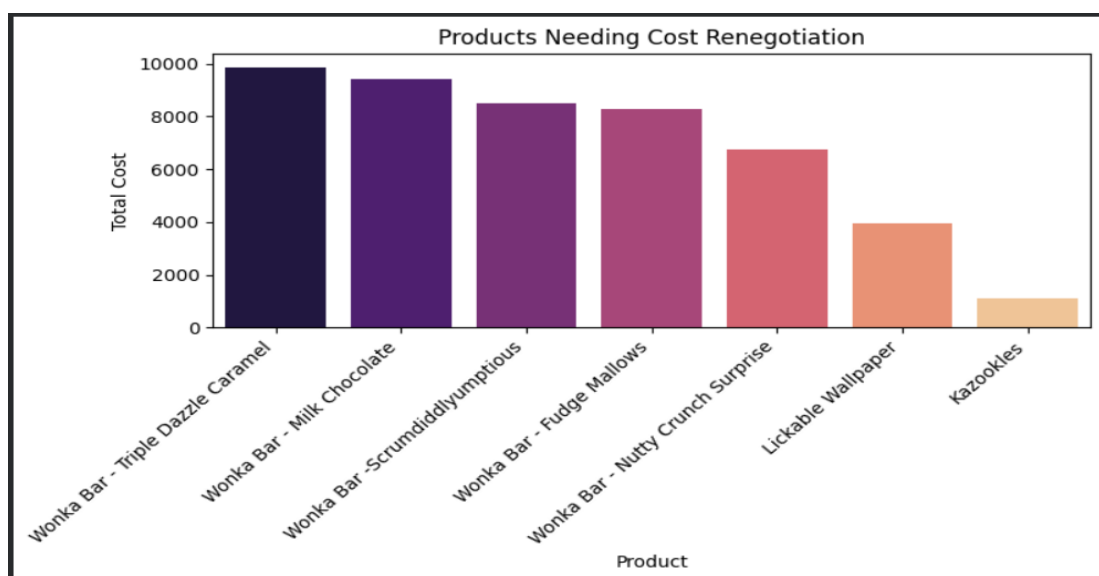
The regional revenue analysis highlights a significant concentration of sales within a small number of states. California emerges as the largest revenue-generating market, substantially outperforming all other regions, followed by New York and Texas. Together, these three states contribute a considerable share of total company revenue, indicating a strong dependence on a limited geographic footprint. Revenue declines noticeably after the top-performing states, suggesting that several regions contribute only marginally to overall sales. While this concentration provides strong revenue streams from established markets, it also creates potential business risks associated with regional economic fluctuations, competitive pressures, or supply chain disruptions. The findings suggest an opportunity to strengthen market penetration in lower-performing regions while maintaining customer retention and growth initiatives in the company's highest-value markets.

9. Cost vs Sales Analysis



The Cost vs Sales analysis reveals a strong positive relationship between product cost and revenue generation, indicating that products with higher investment levels generally achieve higher sales performance. Most products cluster within the lower cost range, while a small group of high-cost products generates substantially greater sales volumes. The variation in bubble size and gross margin percentages suggests that higher costs do not necessarily guarantee superior profitability, emphasizing the importance of balancing sales growth with margin performance. The findings help identify products that efficiently convert cost investments into revenue and support data-driven pricing, sourcing, and product portfolio decisions.

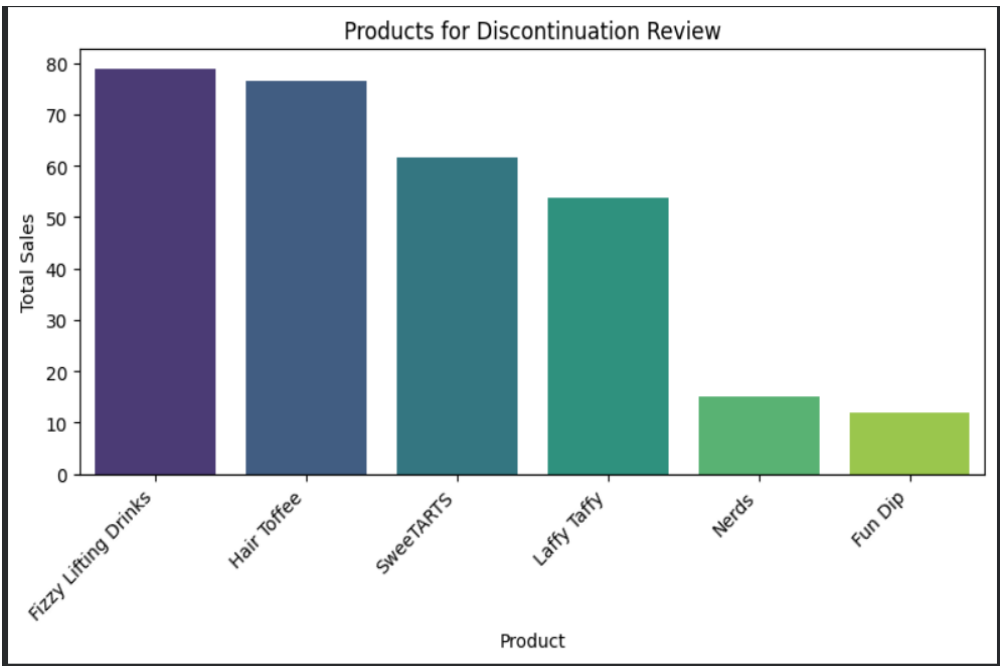
10. Products Needing Cost Renegotiation Analysis



The cost renegotiation analysis identifies products with the highest procurement or production costs that may benefit from supplier negotiations or cost optimization

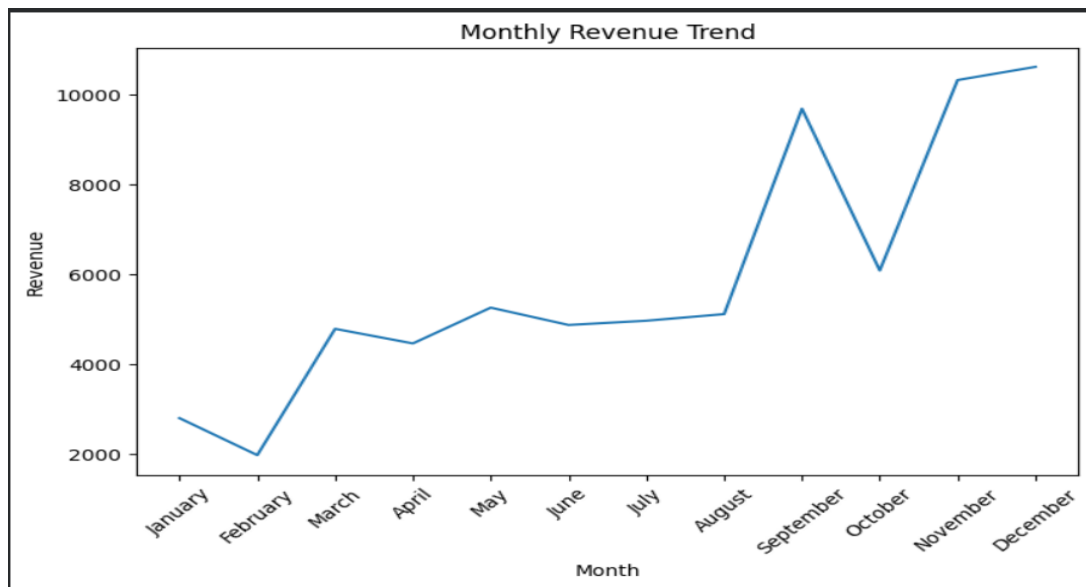
initiatives. The Wonka Bar product line accounts for the majority of high-cost products, with Triple Dazzle Caramel, Milk Chocolate, Scrumdiddlyumptious, and Fudge Mallows showing the largest cost values. While these products generate strong sales and profits, their substantial cost structure presents opportunities for margin improvement through better supplier pricing, bulk purchasing agreements, or operational efficiencies. Additionally, products such as Kazookles exhibit relatively high costs compared to their profitability, indicating potential inefficiencies that require closer review. The findings support strategic cost management efforts aimed at improving profitability without negatively impacting revenue performance.

11.Product Portfolio Optimization Analysis



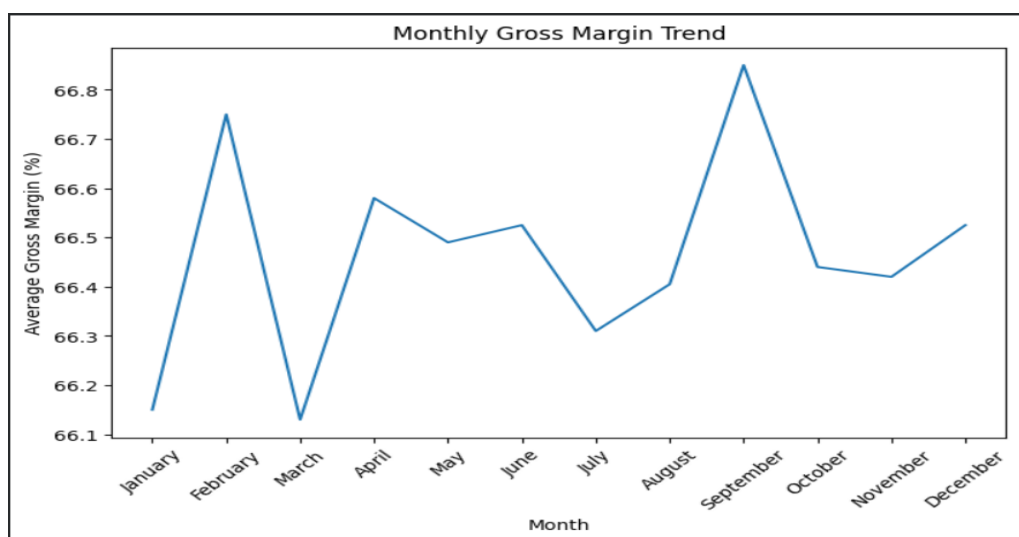
The discontinuation review identifies products with both below-average sales and below-average gross profit, making them potential candidates for portfolio optimization. Products such as Fizzy Lifting Drinks, Hair Toffee, SweetTARTS, Laffy Taffy, Nerds, and Fun Dip contribute relatively little to overall revenue and profitability compared to the company's leading products. While some of these products maintain reasonable gross margins, their limited sales volume restricts their overall business impact. These products should be evaluated for strategic relevance, customer demand, inventory carrying costs, and growth potential before making discontinuation decisions. The findings support a more focused product portfolio by prioritizing resources toward high-performing products that drive the majority of revenue and profit.

12. Time-Based Revenue Analysis – Monthly Revenue Trend



The monthly revenue trend reveals a clear increase in sales performance throughout the year, with revenue accelerating significantly during the final quarter. Revenue remained relatively stable between March and August, averaging around \$4,500–\$5,200 per month, before experiencing a sharp surge in September. Although revenue temporarily declined in October, it rebounded strongly in November and reached its highest level in December. This pattern suggests the presence of seasonal demand, likely driven by holiday purchasing and increased consumer spending during the year-end period. The findings indicate that inventory planning, production scheduling, and marketing campaigns should be intensified ahead of the fourth quarter to capitalize on peak demand and maximize revenue opportunities.

13. Monthly Average Margin Analysis



The monthly gross margin remained remarkably stable throughout the year, fluctuating within a narrow range of approximately 66%–67%. The highest margin was recorded in September, while the lowest occurred in March, but the overall variation was minimal. The calculated margin volatility of only 0.32% confirms that Nassau Candy maintained consistent profitability despite significant changes in monthly revenue. This indicates strong pricing discipline, effective cost control, and a stable product mix across the year. Since profit margins remained steady while revenue increased substantially during peak months, future profitability growth is more likely to come from expanding sales volume rather than relying on margin improvements.

8. Recommendations

Based on the analytical findings, the following recommendations are proposed:

- Increase inventory allocation and production planning for the **Chocolate Division**, which consistently generates the highest revenue, profit, and gross margin.
- Reduce dependency on a small group of products by expanding and promoting underperforming product lines to create a more balanced revenue portfolio.
- Prioritize marketing investments on the top-performing **Wonka product range**, as a small percentage of products contribute the majority of total revenue and profit.
- Conduct supplier negotiations and cost optimization initiatives for products with high cost structures, particularly those identified in the cost renegotiation analysis.
- Review low-sales and low-profit products regularly to determine whether repositioning, repricing, bundling, or discontinuation would improve overall profitability.
- Develop targeted sales strategies for high-potential states such as California, New York, and Texas while identifying opportunities to increase market penetration in lower-performing regions.
- Strengthen demand forecasting and inventory management ahead of peak seasonal periods, particularly during the fourth quarter when revenue reaches its highest levels.
- Continue monitoring gross margin performance across divisions and products to maintain the company's strong margin stability and profitability levels.

These recommendations can improve revenue growth, profitability, product portfolio efficiency, and long-term business sustainability for Nassau Candy Distributor.

9. Future Scope

Future enhancements may include integrating predictive analytics and advanced business intelligence capabilities to improve revenue forecasting, product portfolio management, and operational planning.

Potential improvements include:

- Product demand forecasting models
- Revenue and profit prediction using machine learning
- Customer segmentation and purchasing behavior analysis
- Dynamic pricing optimization strategies
- Regional sales forecasting and market expansion analysis
- Inventory optimization and stock replenishment planning
- Supplier performance and procurement analytics
- Real-time business performance monitoring dashboards
- Advanced profitability modeling at customer and product levels
- Automated alerts for declining sales, margins, or product performance

These capabilities can further improve strategic decision-making, profitability management, inventory efficiency, and long-term growth planning for Nassau Candy Distributor.

9. Conclusion

This project successfully developed a comprehensive profitability and revenue performance analysis for Nassau Candy Distributor using modern data analytics and business intelligence techniques.

The analysis identified key revenue drivers, high-performing divisions, profit concentration patterns, regional sales dependencies, cost management opportunities, and seasonal revenue trends. Findings revealed that the **Chocolate Division** is the strongest contributor to both revenue and profitability, while a relatively small percentage of products generate the majority of total revenue and profit, highlighting a significant concentration risk.

The study also demonstrated that gross margins remain highly stable throughout the year despite fluctuations in sales volume, indicating strong pricing discipline and effective cost control. Additionally, regional analysis showed substantial dependence on a few major states, while product portfolio analysis identified opportunities for cost optimization and product rationalization.

Overall, the project transformed raw sales and profitability data into actionable business insights, supporting data-driven decision-making in product management, inventory planning, supplier negotiations, regional expansion, and profitability optimization. The findings provide a strong foundation for improving operational efficiency, reducing business risk, and driving sustainable revenue growth for Nassau Candy Distributor.